Little Lab Coats

☀️ The Sun — Our Closest Star

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 1 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only

⚠️ Sun Safety — Read Before Starting!

Never look directly at the sun — not even for a split second. The sun's light is so powerful it can permanently damage your eyes. This lesson uses *shadows* as a safe way to study the sun — we never look at the sun itself. Please review this rule with your child before going outside.

- Always keep your back to the sun when doing shadow observations
- If you feel the urge to look up — look at the ground instead
- Wear sunscreen and a hat on sunny days

The sun, Earth, and its shadow — a system that creates our days and nights

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

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By the end of this lesson, students will

- *Describe* the observ
- *Explain* why the s
- *Connect* shadow length and direction to the sun's

🌟 The Big Science Idea

The sun appears to move across the sky

every morning, stay the same

Earth rotates (spins on its axis) once every 24 hours. As we spin, the sun appears to travel across the sky — just like when you spin around in a circle and the trees around you seem to move even though they're standing still. We call this apparent motion.

Because of the sun's position in the sky, shadows change throughout the day. In the morning and afternoon, shadows are long. At midday (when the sun is highest), shadows are shortest.

> 💡 **Key vocabulary:** sun, rotate, orbit, shadow, east, west, midday, apparent motion

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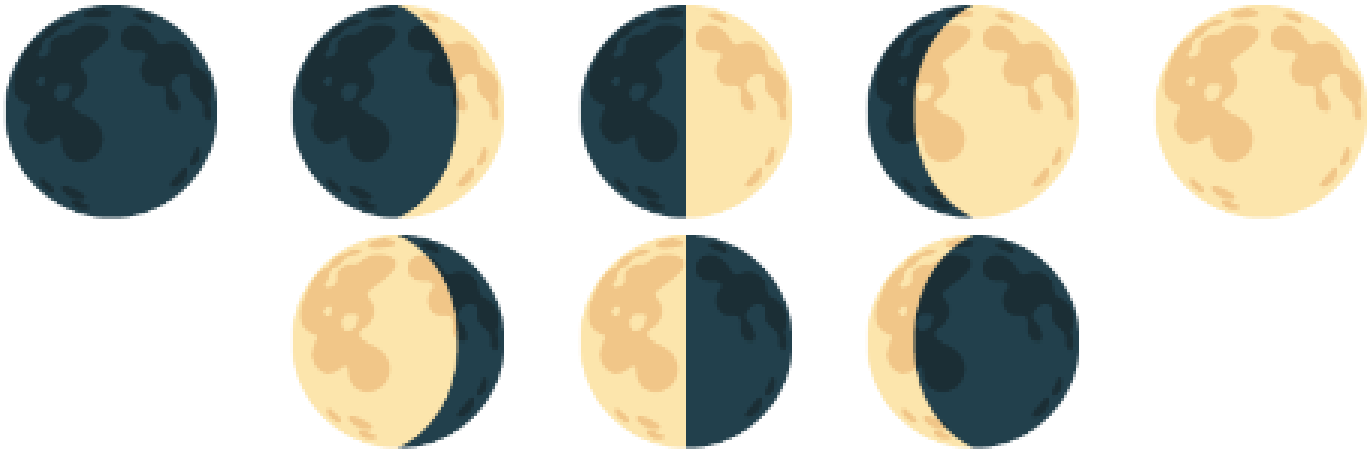
🌙 The Moon — Earth's Natural Satellite

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 2 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes + ongoing nightly observations 🔬 **Subject:** Earth Science — Sun, Moon & Stars
🏠 **Materials:** Household only



The moon moves through a full cycle of phases every 29.5 days

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Observe* and *record* the moon's apparent shape over several days
- *Describe* the pattern of moon phases and predict what comes next
- *Explain* why the moon appears to change shape (we see different portions of the lit side)

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🔬 The Big Science Idea

The moon seems to cha[illegible] shape). What changes is *how much of the lit side we can see* from Earth.

The moon orbits (travels around) Earth [illegible] moon — but as the moon moves around Earth, our viewing angle changes. Sometimes we see most of the lit half (full moon), sometimes we see just a sliver (crescent), and sometimes the lit side faces entirely away from us [illegible]

💡 **Key vocabulary:** moon, phase, orbit, satellite, reflect, crescent, quarter, gibbous, full moon, new moon, cycle

New Moon

Not visible — lit side faces away from Earth

Crescent

A thin curved sliver of light

First Quarter

Half of the moon lit up

Gibbous

More than half lit up

Full Moon

Whole lit side faces Earth — brightest!

Gibbous

Still more than half lit, fading

Last Quarter

Half lit again — other half this time

Crescent

Thin sliver before new moon returns

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⭐ Stars — Suns Very Far Away

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 3 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only

Every star you see at night is a sun — just unimaginably far away

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Explain* why stars are only visible at night (and why the sun makes them invisible during the day)
- *Describe* observable patterns of stars in the sky, including recognizable groupings
- *Identify* the Big Dipper and three other common star patterns using picture cards and a simple star chart

🔬 The Big Science Idea

Here's a surprising fact: stars are shining in the sky *right now* — even during the day! We just can't see them because our sun is so bright it outshines everything else. When the sun sets and the sky darkens, the become visible again — it's like turning off a bright light in a room and suddenly being able to see the stars painted on the ceiling.

Every single star you see is a sun — only look tiny and faint because they are *unimaginably far away*. The closest star besides our sun (Proxima Centauri) is so far that traveling to it at the speed of an airplane would take about 4 mill

Even though stars seem g Dipper is one of the most reliable and easiest to find — like a ladle or a dip

Key vocabulary: star, constellation, asterism, Big Dipper, light-year, stable, atmosphere, night sky

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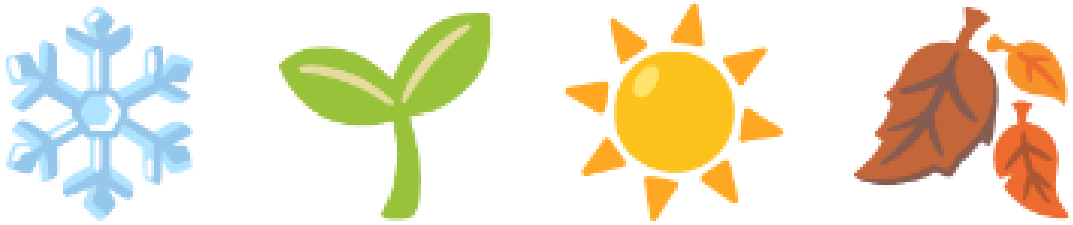
📅 How Much Daylight? Seasons and the Sun

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 4 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only

⚠️ Sun Safety Reminder

This lesson involves observing daylight patterns — not looking at the sun. As always, **never look directly at the sun.** Daylight observations mean noticing when it's light or dark outside — keep eyes away from the sun itself.



Winter, Spring, Summer, Fall — daylight hours change dramatically across the seasons

📋 **NGSS Standard: 1-ESS1-2** — Make observations at different times of year to relate the amount of daylight to the time of year.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Compare* the hours of daylight across the four seasons
- *Explain* the pattern: more daylight in summer, less in winter
- *Identify* Earth's tilt (not distance from the sun) as the cause of seasons
- *Create* a bar graph showing seasonal daylight changes

🔬 The Big Scientific Idea

Have you ever noticed that summer days feel longer than winter days? The number of daylight hours changes throughout the year in a predictable pattern.

In summer, the sun rises early, climbs high in the sky, and sets late. In winter, the sun rises later, stays low in the sky, and sets early — giving us less daylight.

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Here's the surprising part: this has *nothing* to do with how close Earth is to the sun! Earth is actually slightly *closer* to the sun in January (winter in the Northern Hemisphere) than in July (summer). The cause of seasons is Earth's *tilt*.

Earth spins at a tilt of about 23.5 degrees. In summer, our hemisphere is tilted *toward* the sun — days are longer and sunlight hits us more directly (more energy, more warmth). In winter, our hemisphere is tilted *away* — days are shorter and sunlight hits us at a glancing angle (less energy, less warmth).

> 💡 **Key vocabulary:** season, daylight, sunrise, sunset, tilt, hemisphere, pattern, summer solstice, winter solstice, equinox

> ☀️ **Sample Daylight Hours (Denver, CO — approximate):**
> Winter solstice (Dec 21): ~9 hours 21 minutes of daylight
> Spring equinox (Mar 20): ~12 hours 10 minutes of daylight
> Summer solstice (Jun 21): ~14 hours 58 minutes of daylight
> Fall equinox (Sep 22): ~12 hours 10 minutes of daylight
> *Look up your location's actual times at* ***timeanddate.com/sun***

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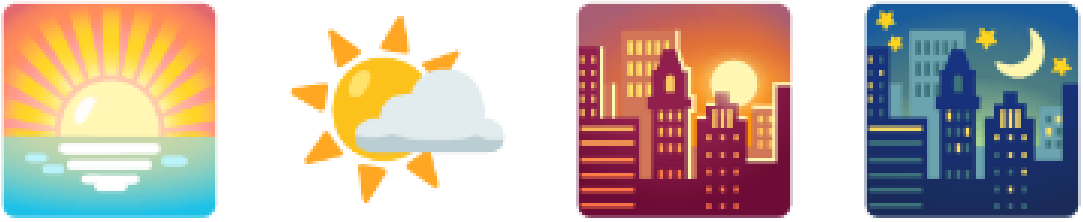
🌅 The Sky Changes — Morning, Day & Night

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 5 of 6

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes + brief observations across one day 🔬 **Subject:** Earth Science — Sun, Moon & Stars
🏠 **Materials:** Household only

⚠️ Sun Safety Reminder

When observing the sky at sunrise or during the day, **always look at the sky around the sun — not directly at the sun itself.** You can observe sky colors, clouds, and position of light without ever looking at the sun directly. Keep this rule all day!



Sunrise, midday, sunset, night — the sky transforms completely across a single day

📋 **NGSS Standard: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Observe* and *record* how the sky's appearance changes from morning to night
- *Identify* patterns in sky color, light, cloud and object visibility at different times of day
- *Explain* why the sky is blue during the day and red/orange at sunrise and sunset

🔭 The Big Scientific Idea

The sky is never the same twice — it changes from morning to night. But it follows a *predictable pattern* that repeats every single day!

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🌅 **Sunrise:** The sky glows orange, pink, and red near the horizon. The sun rises in the east. Stars fade away as the sky brightens.
🌤️ **Midday:** The sky is bright blue. The sun is highest. Shadows are shortest. No stars visible.
🌇 **Sunset:** The sky turns orange, red, and purple again near the horizon. The sun sets in the west.
🌃 **Night:** The sky is dark. Stars appear as Earth rotates away from the sun. The moon may be visible.

Why does the sky change color? It's all about how sunlight travels through Earth's atmosphere. Sunlight contains *all the colors of the rainbow*. When sunlight enters the atmosphere, gas molecules scatter the shorter (blue) wavelengths in all directions — this scattered blue light is what makes the sky look blue during the day.

At sunrise and sunset, sunlight has to travel through much more atmosphere to reach your eyes (because the sun is near the horizon). By the time it arrives, most of the blue light has already scattered away. What's left are the longer red and orange wavelengths — giving us those gorgeous pink and orange skies!

💡 **Key vocabulary:** atmosphere, scatter, wavelength, Rayleigh scattering, sunrise, sunset, observation, pattern

🌈 Why the Sky Changes Color — A Visual

Sunrise 🌅	Midday 🌤️	Sunset 🌇	Night 🌃

At sunrise and sunset, light travels through more atmosphere → blue scatters away → orange/red remains. At midday, light travels through less atmosphere → blue dominates.

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🔭 Patterns in the Sky

Earth Science — Sun, Moon & Stars | Unit 1, Lesson 6 of 6 ⭐ Final Lesson

📚 **Grade:** 1st Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Earth Science — Sun, Moon & Stars 🏠 **Materials:** Household only

The sun, moon, and stars — three cosmic objects with patterns scientists can predict for centuries

📋 **NGSS Standards: 1-ESS1-1** — Use observations of the sun, moon, and stars to describe patterns that can be predicted. | **1-ESS1-2** — Make observations at different times of year to relate the amount of daylight to the time of year.

🎯 Learning Objectives

By the end of this lesson, students will be able to:

- *Summarize* the observable patterns of the sun, moon, and stars across different time periods
- *Predict* what the sky will look like at a given time based on known patterns
- *Connect* all unit concepts — daily sun motion, moon phases, star visibility, daylight seasons, sky color changes

🔬 What We've Learned: Sky Patterns Summary

Object	Pattern Over a Day	Pattern Over a Month	Pattern Over a Year
☀️ **Sun**	Rises east, arcs across the sky, sets west. Shadow moves and changes length	Same daily pattern, but slowly changes position	Higher in summer (more daylight), lower in winter (less daylight)
🌙 **Moon**	May rise at different times depending on phase	Moves from new to full and back to new	13 times per year

🥼

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⭐ **Stars**	Not visible in daytime (outshone by sun). Appear at dusk, move slowly through night.	Constellations shift position as Earth orbits, but patterns stay the same	Different constellations visible in different seasons (Earth's orbital position changes which stars face our night side)

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